CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2016 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Byte-code in JAVA? Explain.

Definition: The intermediate machine instruction set generated by `javac`, executed by the JVM to provide platform independence.

Q2 (3M) What is Garbage Collection in Java?

Definition: Automatic memory reclamation mechanism of the JVM destroying unreferenced heap objects.

Q3 (3M) Differentiate between Class and Interface.

Differences: A class contains full implementation and state; an interface contains method specifications and constants.

Q4 (3M) What is an Applet?

Definition: A small Java program designed to be embedded in an HTML web page and executed inside a web browser.

Q5 (3M) What is Thread in Java?

Definition: A lightweight sub-process representing an independent execution path in a program.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) What do you mean by Constructor? Explain Constructor Overloading.

Constructor Overloading: Defining multiple constructors in a single class with different parameter lists.

Q7 (7.5M) Explain Packages in Java and how to create user-defined packages.

Packages: Directory-based namespaces created with `package` statement.

Q8 (7.5M) Explain Exception Handling in Java with try, catch, and finally.

Handling: Graceful recovery from runtime program faults.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain Applet Life Cycle in detail with an Applet code to draw shapes.

Applet Shapes Code: Complete applet drawing rectangles, lines, and ovals using `Graphics`.

Q10 (15M) Explain Multithreading and Thread Synchronization with examples.

Multithreading & Synchronization: Producer-Consumer problem and synchronized locks.

Q11 (15M) Explain JDBC Architecture and steps to connect Java to a database.

JDBC: Complete connectivity steps and SQL operations.

Q12 (15M) **Explain Servlets, HttpServlet, and its life cycle methods.**

Servlets: Complete server-side Java web request dispatching.

Q13 (15M) **Explain Client-Server Socket Programming in Java with complete program.**

Socket Programming: `ServerSocket` and `Socket` communication code over TCP.
